

ALU HW1說明

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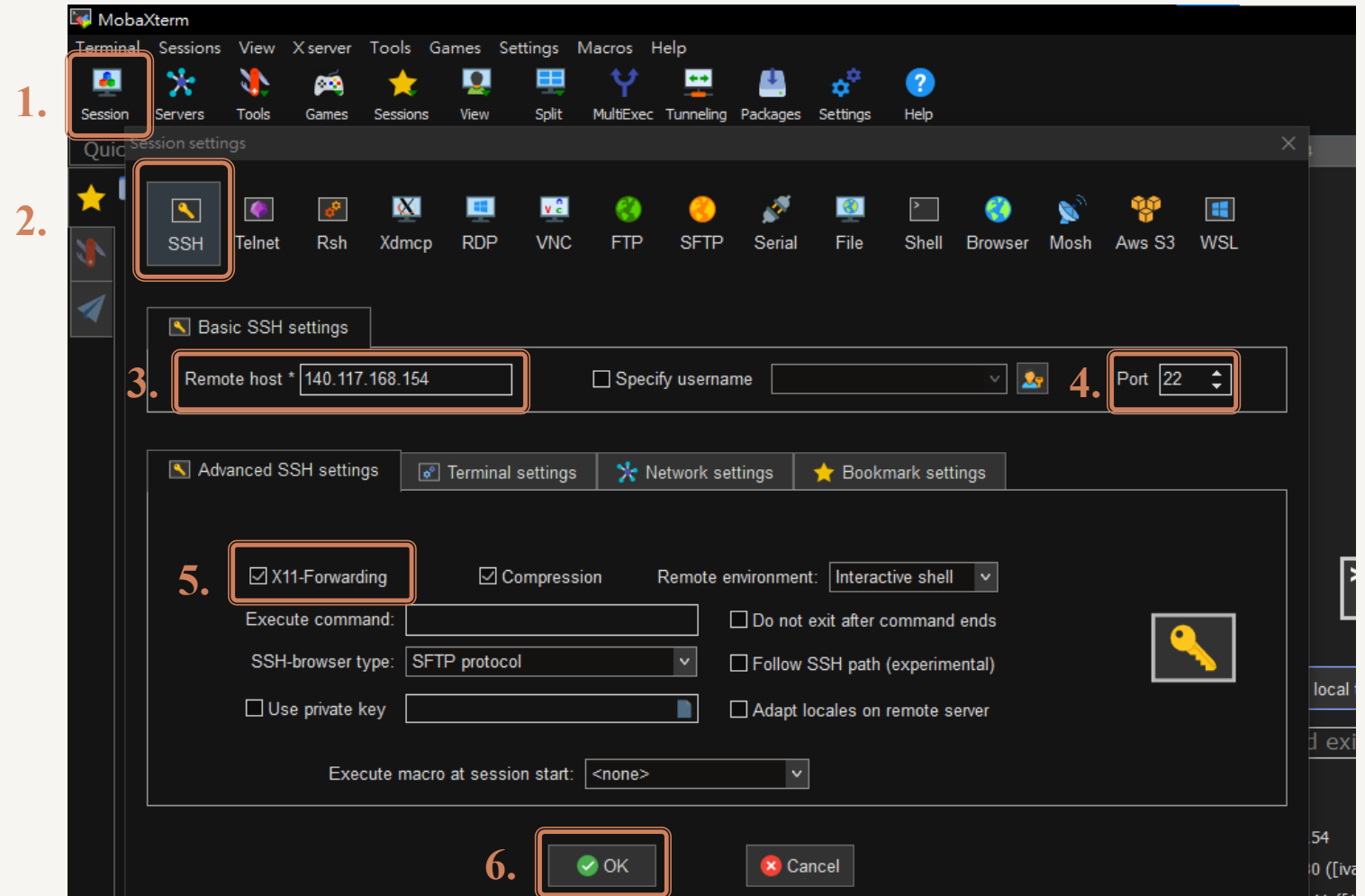
2026.09.16

Outline

- EDA Server
- EDA Tools
 - VCS
 - Design Compiler
 - nWave, Verdi
- HW1目錄建立方式與流程
- 檔案、top module命名限制
- test bench
- HW1 補充說明：pipeline
- Report
- 評分標準與繳交方式

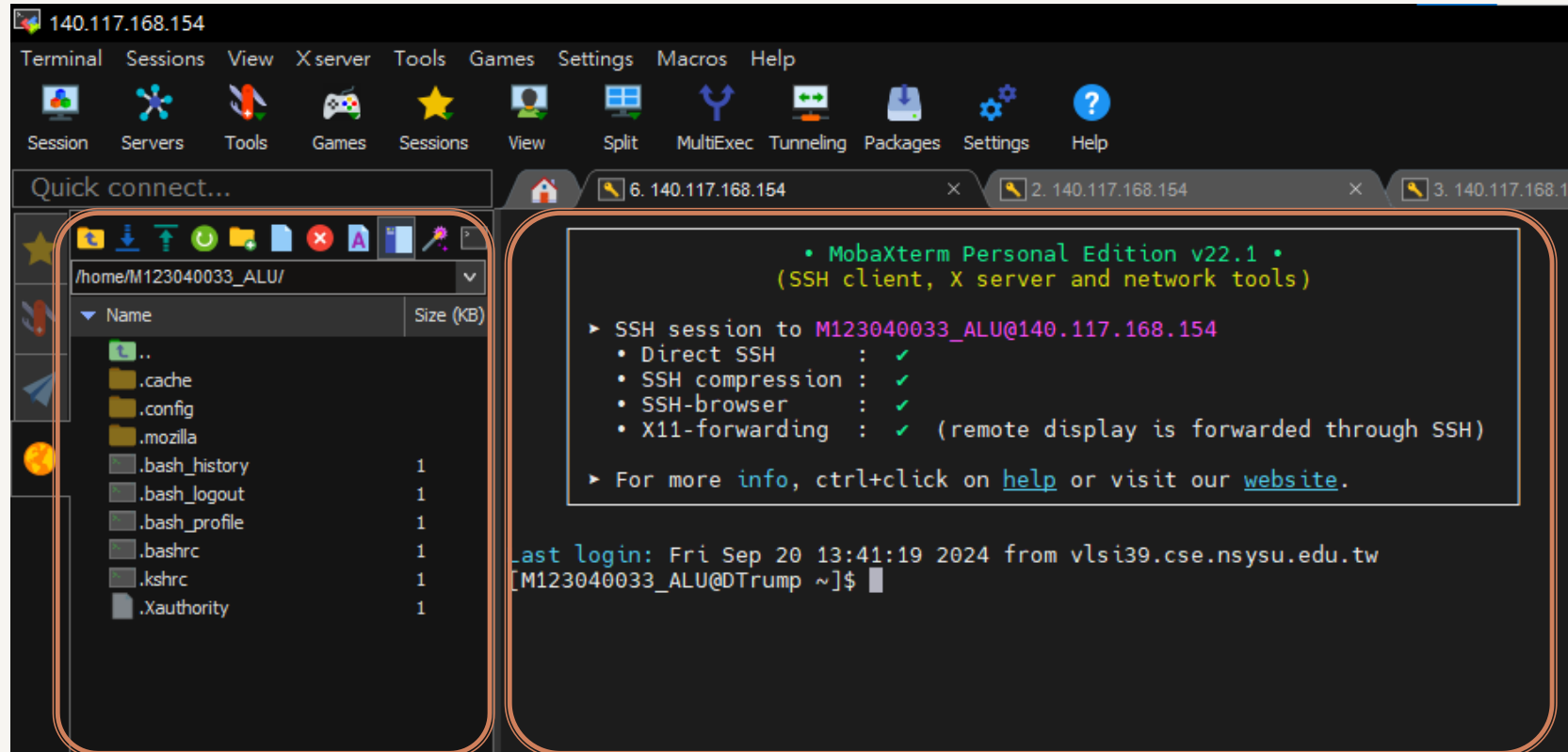
EDA Server (1/3)

請使用遠端程式 MobaXterm
並用SSH連入伺服器



EDA Server (2/3)

登入成功

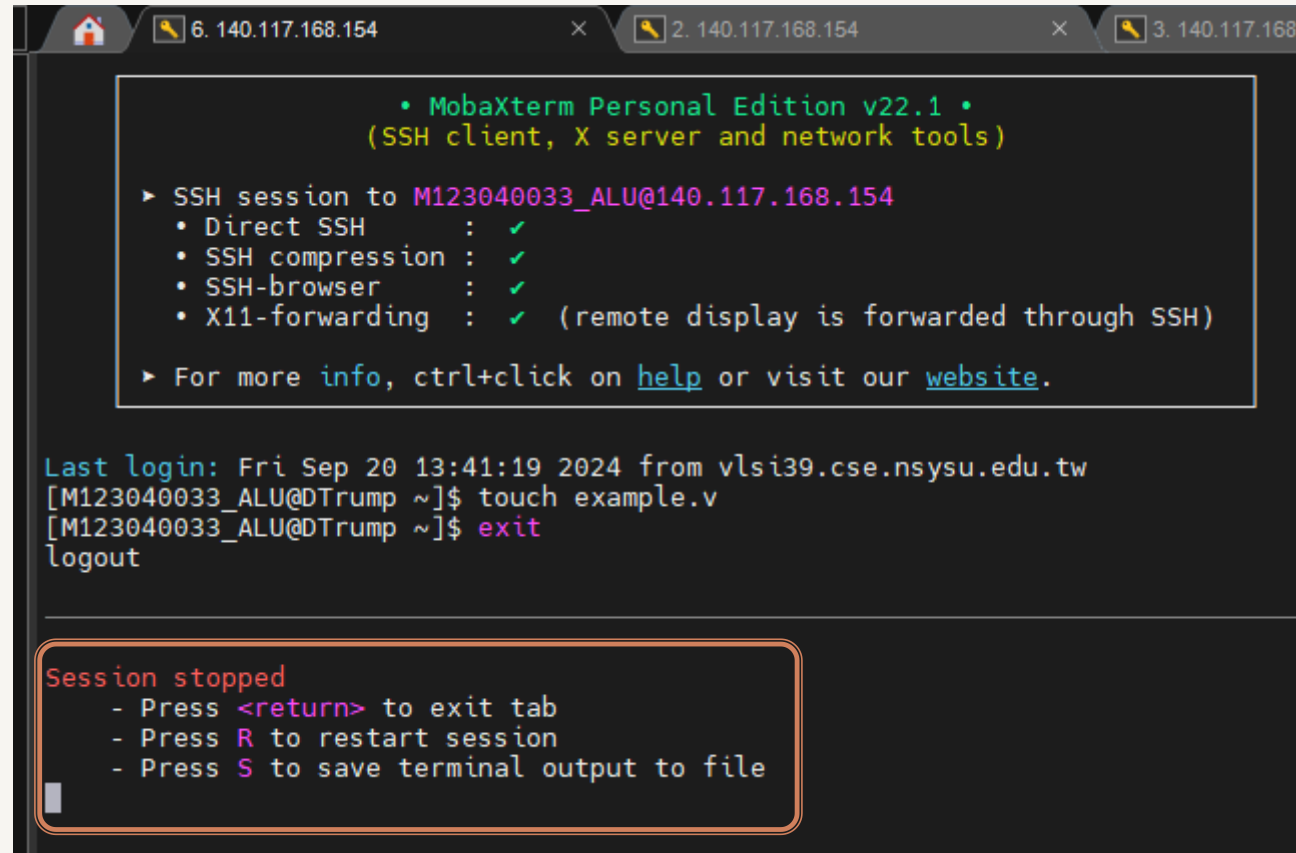


檔案

終端機

EDA Server (3/3)

使用完畢後請務必輸入 `exit` / (Ctrl + D) 關閉連線以避免佔用伺服器資源



```
• MobaXterm Personal Edition v22.1 •
(SSH client, X server and network tools)

▶ SSH session to M123040033_ALU@140.117.168.154
  • Direct SSH      : ✓
  • SSH compression : ✓
  • SSH-browser     : ✓
  • X11-forwarding  : ✓ (remote display is forwarded through SSH)

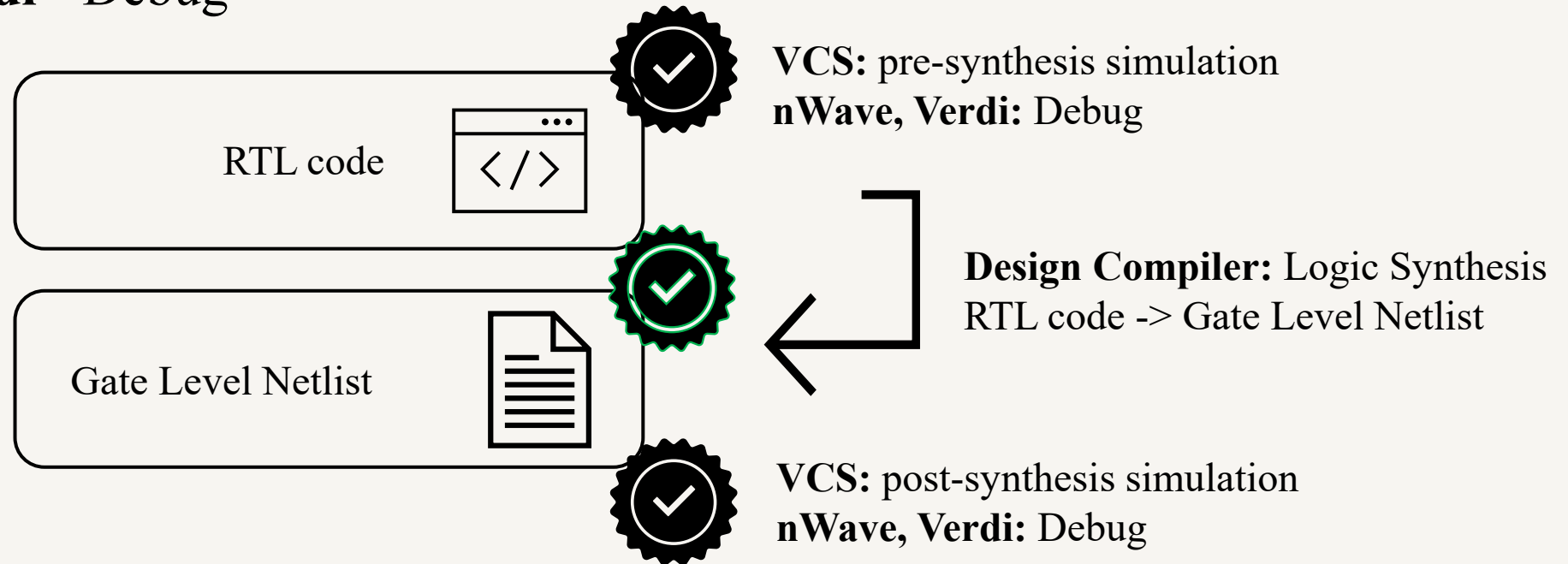
▶ For more info, ctrl+click on help or visit our website.

Last login: Fri Sep 20 13:41:19 2024 from vlsi39.cse.nsysu.edu.tw
[M123040033_ALU@DTrump ~]$ touch example.v
[M123040033_ALU@DTrump ~]$ exit
logout

Session stopped
- Press <return> to exit tab
- Press R to restart session
- Press S to save terminal output to file
```

EDA Tools

- **VCS** - Simulation
- **Design Compiler** – Synthesis
- **nWave, Verdi** - Debug



VCS simulation

- 利用 Testbench 提供測試訊號
- 模擬電路訊號與運作狀況
- 輸入檔案
 - Testbench
 - Your Design
- 輸出檔案
 - .fsdb Waveform file

VCS pre-synthesis simulation (1/2)

1. 輸入 tcsh
2. 編輯 pre_sim.sh

```
[m123040033@DTrump pre_sim]$ tcsh
set vcs version: 2023.12 (default)
set verdi version: 2023.12 (default)
set synthesis version: 2023.12/ (default)
set lc version: 2023.12/ (default)
set formality version: 2023.12/ (default)
set primetime version: 2023.12/ (default)
set INNOVUS version: INNOVUS_21.17.000 (default)
```

testbench

Your Design

Output Waveform

```
$ sim.sh  X  Workspace Trust
C: > Users > ivan > AppData > Roaming > MobaXterm > slash > RemoteFiles > 133708_2_2 > $ sim.sh
1  #!/bin/tcsh
2
3  vcs -R \
4  /home/m123040033/HDL/pre_sim/TB.v \
5  /home/m123040033/HDL/rtl/adder_behavior_reg.v\
6  /home/m123040033/HDL/rtl/adder_behavior.v\
7  /home/m123040033/HDL/rtl/adder_dataflow_reg.v\
8  /home/m123040033/HDL/rtl/adder_dataflow.v\
9  /home/m123040033/HDL/rtl/adder_structure_reg.v\
10 /home/m123040033/HDL/rtl/adder_structure.v\
11 /home/m123040033/HDL/rtl/D_FF_32.v\
12 /home/m123040033/HDL/rtl/D_FF.v\
13 /home/m123040033/HDL/rtl/FullAdder.v \
14 +full64 \
15 +access+r +vcs+fsdbon +fsdb+mda +fsdbfile+6adder.fsdb
16
```

建議

1. 使用絕對路徑
2. 自己打路徑的內容

VCS pre-synthesis simulation (2/2)

3. source pre_sim.sh

```
[m123040033@Elden pre_sim]$ source sim.sh
*** Using c compiler gcc instead of cc ...
          Chronologic VCS (TM)
          Version V-2023.12 -- Fri Sep 20 16:06:47 2024
```

```
          Copyright (c) 1991 - 2023 Synopsys, Inc.
This software and the associated documentation are proprietary to Synopsys,
Inc. This software may only be used in accordance with the terms and conditions
of a written license agreement with Synopsys, Inc. All other use, reproduction,
or distribution of this software is strictly prohibited. Licensed Products
communicate with Synopsys servers for the purpose of providing software
updates, detecting software piracy and verifying that customers are using
Licensed Products in conformity with the applicable License Key for such
Licensed Products. Synopsys will use information gathered in connection with
this process to deliver software updates and pursue software pirates and
infringers.
```

```
Inclusivity & Diversity - Visit SolvNetPlus to read the "Synopsys Statement on
Inclusivity and Diversity" (Refer to article 000036315 at
https://solvnetplus.synopsys.com)
```

```
Parsing design file '/home/m123040033/HDL/pre_sim/TB.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_behavior_reg.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_behavior.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_dataflow_reg.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_dataflow.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_structure_reg.v'
Parsing design file '/home/m123040033/HDL/rtl/adder_structure.v'
Parsing design file '/home/m123040033/HDL/rtl/D_FF_32.v'
Parsing design file '/home/m123040033/HDL/rtl/D_FF.v'
Parsing design file '/home/m123040033/HDL/rtl/FullAdder.v'
Top Level Modules:
```

```
    HW1_TB
TimeScale is 1 ns / 100 ps
Starting vcs inline pass...
```

```
3 modules and 0 UDP read.
recompiling module HW1_TB
recompiling module adder_structure
recompiling module FA
All of 3 modules done
```

```
3 modules and 0 UDP read.
recompiling module HW1_TB
recompiling module adder_structure
recompiling module FA
All of 3 modules done
rm -f _cuarc*.so _csrc*.so pre_vcsobj_*.so share_vcsobj_*.so
if [ -x ../simv ]; then chmod a-x ../simv; fi
g++ -o ../simv -m32 -m32 -rdynamic -Wl,-rpath='$_ORIGIN'/simv.daidir -Wl,-rpath=../simv.da
idir -Wl,-rpath=/usr/cad/synopsys/vcs/2023.12/linux/lib -L/usr/cad/synopsys/vcs/2023.12/linux/l
ib -Wl,-rpath-link=../objs/amcQw_d.o _4401_archive_1.so _prev_archive_1.so SIM_l.o
rmapats_mop.o rmapats.o rmar.o rmar_nd.o rmar_llvm_0_1.o rmar_llvm_0_0.o -lvirsim
-lerrorinf -lsnpsmalloc -lvfs -lvcsnew -lsimprofile -luclinate /usr/cad/synopsys/vcs/2023.
12/linux/lib/vcs_tls.o -Wl,-whole-archive -lvcsucli -Wl,-no-whole-archive _vcs_pli
_stub.o /usr/cad/synopsys/vcs/2023.12/linux/lib/vcs_save_restore_new.o /usr/cad/synopsys/ver
di/2023.12/share/PLI/VCS/LINUX/pli.a /usr/cad/synopsys/vcs/2023.12/linux/lib/ctype-stubs_32.a -
ldl -lc -lm -lpthread -ldl
../simv up to date
Info: [VCS_SAVE_RESTORE_INFO] ASLR (Address Space Layout Randomization) is detected on the mach
ine. To enable $save functionality, ASLR will be switched off and simv re-executed.
Please use '-no_save' simv switch to avoid re-execution or '-suppress=ASLR_DETECTED_INFO' to su
ppress this message.
Chronologic VCS simulator copyright 1991-2023
Contains Synopsys proprietary information.
Compiler version V-2023.12; Runtime version V-2023.12; Sep 20 16:07 2024
*Verdi* Loading libsscore_vcs202312.so
FSDB Dumper for VCS, Release Verdi_V-2023.12, Linux, 11/15/2023
(C) 1996 - 2023 by Synopsys, Inc.
*Verdi* FSDB WARNING: The FSDB file already exists. Overwriting the FSDB file may crash the pro
grams that are using this file.
*Verdi* : Create FSDB file '6adder.fsdb'
*Verdi* : Begin traversing the scopes, layer (0).
*Verdi* : Enable +mda dumping.
*Verdi* : End of traversing.
$finish called from file "/home/m123040033/HDL/pre_sim/TB.v", line 35.
$finish at simulation time 1000
V C S S i m u l a t i o n R e p o r t
Time: 100000 ps
CPU Time: 0.660 seconds; Data structure size: 0.0Mb
Fri Sep 20 16:07:10 2024
CPU time: .558 seconds to compile + .499 seconds to elab + .412 seconds to link + .722 seconds
in simulation
```

Design Compiler (1/10)

輸入 `dcnxt_shell -f dc.tcl` 開始執行 tcl 檔、進行合成

```
[m103040049@devotion ~/ALU_HW1]$ dc_shell -f dc.tcl

      Design Compiler Graphical
      DC Ultra (TM)
      DFTMAX (TM)
      Power Compiler (TM)
      DesignWare (R)
      DC Expert (TM)
      Design Vision (TM)
      HDL Compiler (TM)
      VHDL Compiler (TM)
      DFT Compiler
      Design Compiler(R)

      Version 0-2018.06 for linux64 - May 21, 2018

      Copyright (c) 1988 - 2018 Synopsys, Inc.
      This software and the associated documentation are proprietary to Synopsys,
      Inc. This software may only be used in accordance with the terms and conditions
      of a written license agreement with Synopsys, Inc. All other use, reproduction,
      or distribution of this software is strictly prohibited.
      Initializing...
      set Company          "VLSI5015"
      VLSI5015
      set Designer         "default"
      default
      #設定40nm製程路徑
      set search_path      "/EDA Tools/CBDK/CBDK TSMC40 Arm f2.0/CIC/SynopsysDC/db/sc9 b
```

所需檔案：

- RTL code (你的程式碼，不包含testbench)
- 執行 Design compiler 之腳本 (.tcl檔)
- TSMC ADFP Cell Library File (伺服器內部)
- EDA Tools (伺服器內部)

Design Compiler (2/10)

腳本說明：dc.tcl

```
4  #設定ADFP(16nm)製程路徑
5  set search_path      "/cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/CCS/ $search_path"
6  #設定ADFP(16nm)製程路徑檔，如果有memory compiler的檔案db檔的路徑，記得在這邊設定
7  set target_library    "N16ADFP_StdCellss0p72vm40c_ccs.db N16ADFP_StdCellff0p88v125c_ccs.db"
8
9  set link_library      "* $target_library dw_foundation.sldb"
10 set symbol_library    "tsmc040.sdb generic.sdb"
11 set synthetic_library "dw_foundation.sldb"
12 set hdlin_translate_off_skip_text "TRUE"
13 set edifout_netlist_only "TRUE"
14 set verilout_no_tri true
15 set hdlin_enable_presto_for_vhdl "TRUE"
16 set sh_enable_line_editing true
17 set sh_line_editing_mode emacs
18 history keep 100
19 alias h history
```

Design Compiler (3/10)

- 修改檔案存放位置、top module name 及 clock

```
21 #Path_Top:Verilog放置的位置
22 #Path_Syn:合成後report.txt檔案要放置的根位置，需自行在目錄下創建名為dc_out_file之資料夾
23 #Dump_file_name:合成後產生檔案之名字
24 set Path_Top      "/"
25 set Path_Syn      "../structure_reg/"
26 set Dump_file_name "adder_structure_reg"
27 #設定Top module 名稱，需跟自行設計之電路的top module name相同
28 set Top           "adder_structure_reg"
29 #Specify Clock, clock名需和top module中clk port相同
30 set Clk_pin       "clk"
31 set Clk_period    "0.49"
```

- 將所有要合成的檔案之完整路徑放入 {} 內

```
33 # [ Read Design File ]
34 #如果設計有parameter設計，read_file指定不能用，需使用analyze + elaborate指令並自行更改路徑
35 # read_file -format verilog {/home/m103040049/HDL_HW/multiplier.v}
36 # current_design $Top
37 analyze -format verilog {
38     /home/m123040033/HDL/rtl/adder_structure_reg.v
39     /home/m123040033/HDL/rtl/D_FF.v
40     /home/m123040033/HDL/rtl/D_FF_32.v
41     /home/m123040033/HDL/rtl/FullAdder.v
42 }
43 elaborate $Top
44 current_design $Top
45 #檢查是否讀取成功
46 link
```

Design Compiler (4/10)

- 作業要求合成 delay-optimization, area-optimization, in-between 三種電路，需修改period

```
21  #Path_Top:Verilog放置的位置
22  #Path_Syn:合成後report.txt檔案要放置的根位置，需自行在目錄下創建名為dc_out_file之資料夾
23  #Dump_file_name:合成後產生檔案之名字
24  set Path_Top      "/"
25  set Path_Syn      "../structure_reg/"
26  set Dump_file_name "adder_structure_reg"
27  #設定Top module 名稱，需跟自行設計之電路的top module name相同
28  set Top           "adder_structure_reg"
29  #Specify Clock，clock名需和top module中clk port相同
30  set Clk_pin       "clk"
31  set Clk_period    "0.49"
```

Design Compiler (5/10)

- 先將 period 設一個很大的值 (假設10) 進行合成，合成後 report timing 發現最快可以跑 2，則將 period 改設 2 再合一次，如果 $\text{slack} \geq 0$ (MET) 則 2 即為 **area-optimization** 之 period
- 接著改設 period 為一個很小的值 (假設0) 進行合成，合成後 report timing 發現 $\text{slack} = -0.5$ (Violate)，則將 period 改設 0.5 再合一次，如果 $\text{slack} \geq 0$ (MET) 則 0.5 即為 **delay-optimization** 之 period，讓 slack 盡可能越接近 0
- 最後設 period 為 $(2+0.5)/2=1.25$ 進行合成，假設 $\text{slack} = -0.1$ (Violate) 則設 1.35 再合一次，直到 $\text{slack} \geq 0$ (MET)，最終 period 值即為 **in-between** 電路 之 period

Design Compiler (6/10)

- 如果電路中無 clk 設計，使用 50、51 行，註解 54~57 行

```
49 # [ Setting Clock Constraints, Combinational Circurt USED ]
50 set_max_delay $Clk_period -from [all_inputs] -to [all_outputs]
51 create_clock -name $Clk_pin -period $Clk_period
52
53 # [ Setting Clock Constraints ]
54 # create_clock -name $Clk_pin -period $Clk_period [get_ports $Clk_pin]
55 # set_fix_hold [get_clocks $Clk_pin]
56 # set_dont_touch_network [get_clocks $Clk_pin]
57 # set_ideal_network [get_ports $Clk_pin]
```

- 如果電路中有 clk 設計，使用 54~57 行，註解 50、51 行

```
49 # [ Setting Clock Constraints, Combinational Circurt USED ]
50 # set_max_delay $Clk_period -from [all_inputs] -to [all_outputs]
51 # create_clock -name $Clk_pin -period $Clk_period
52
53 # [ Setting Clock Constraints ]
54 create_clock -name $Clk_pin -period $Clk_period [get_ports $Clk_pin]
55 set_fix_hold [get_clocks $Clk_pin]
56 set_dont_touch_network [get_clocks $Clk_pin]
57 set_ideal_network [get_ports $Clk_pin]
```

Design Compiler (7/10)

- 設定合成時命名規則 (不須修改)
- 輸出 timing、area 和 power report，輸出路徑可至 25 行更改
- 產生合成後 .v 檔、ddc、sdf 和 sdc 檔，產生之名稱可至 26 行更改

```
77 #Change Naming Rule
78 set bus_inference_style {%s[%d]}
79 set bus_naming_style {%s[%d]}
80 set hdlout_internal_busses true
81 change_names -hierarchy -rule verilog
82 define_name_rules name_rule -allowed "A-Z a-z 0-9 _" -max_length 255 -type cell
83 define_name_rules name_rule -allowed "A-Z a-z 0-9 _[]" -max_length 255 -type net
84 define_name_rules name_rule -map {"\\*cell\\*" "cell"}
85 define_name_rules name_rule -case_insensitive
86 change_names -hierarchy -rules name_rule
87 remove_unconnected_ports -blast_buses [get_cells -hierarchical *]
88
89 report_timing -significant_digits 6 -sort_by group
90
91 #Report
92 report_timing -path full -delay max -significant_digits 6 -sort_by group > $Path_Syn/1.timing_report_${Dump_file_name}.txt
93 report_area -hier -nosplit > $Path_Syn/2.area_report_${Dump_file_name}.txt
94 report_power -analysis_effort low > $Path_Syn/3.power_report_${Dump_file_name}.txt
95
96 #Write out
97 write -hierarchy -format ddc -output $Path_Syn/${Dump_file_name}.ddc
98 write -format verilog -hierarchy -output $Path_Syn/${Dump_file_name}.v
99 write_sdf -version 2.1 -context verilog $Path_Syn/${Dump_file_name}.sdf
100 write_sdc $Path_Syn/${Dump_file_name}.sdc
```

Report

2. Area

3. Power

需要 ≥ 0 且有 slack (MET)

Design Compiler (9/10)

Report

1. Timing

2. Area

3. Power

```
*****
Report : area
Design : FLP_adder
Version: 0-2018.06
Date   : Sat Oct  8 17:10:59 2022
*****

Library(s) Used:

    sc9_cln40g_base_rvt_ss_typical_max_0p81v_125c (File: /EDA_Tools/CBDK/CBDK_TSMC40_Arm_f2.0/CIC/SynopsysDC/db/sc9_ba

Number of ports:          1545
Number of nets:           3576
Number of cells:          2029
Number of combinational cells: 1834
Number of sequential cells:  172
Number of macros/black boxes:  0
Number of buf/inv:        409
Number of references:      4

Combinational area:      2369.833178
Buf/Inv area:           218.635199
Noncombinational area:   780.191961
Macro/Black Box area:    0.000000
Net Interconnect area:   undefined (Wire load has zero net area)

Total cell area:         3150.025139
Total area:              undefined

Hierarchical area distribution
-----
```

Combinational Logic (表格 CL 欄位)

Sequential Logic (表格 SL 欄位)

Total Area (表格 Total 欄位)

Design Compiler (10/10)

Report

1. Timing

2. Area

3. Power

Design

Wire Load Model

Library

FLP_adder

Zero

sc9_cln40g_base_rvt_ss_typical_max_0p81v_125c

Global Operating Voltage = 0.81

Power-specific unit information :

Voltage Units = 1V

Capacitance Units = 1.000000pf

Time Units = 1ns

Dynamic Power Units = 1mW (derived from V,C,T units)

Leakage Power Units = 1uW

Cell Internal Power = 287.2819 uW (86%)

Net Switching Power = 46.9000 uW (14%)

Total Dynamic Power = 334.1819 uW (100%)

Dynamic Power (表格 Dynamic 欄位)

Cell Leakage Power = 29.8890 uW

Leakage Power (表格 Leakage 欄位)

Dynamic + Leakage = Total (表格 Total 欄位)

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)	Attrs

io_pad	0.0000	0.0000	0.0000	0.0000	(0.00%)	
memory	0.0000	0.0000	0.0000	0.0000	(0.00%)	
black_box	0.0000	0.0000	0.0000	0.0000	(0.00%)	
clock_network	0.0000	0.0000	0.0000	0.0000	(0.00%)	
register	0.2055	4.9244e-03	7.7363	0.2182	(59.93%)	
sequential	0.0000	0.0000	0.0000	0.0000	(0.00%)	
combinational	8.1768e-02	4.1976e-02	22.1417	0.1459	(40.07%)	

Total	0.2873 mW	4.6900e-02 mW	29.8780 uW	0.3641 mW		

請注意單位

VCS post-synthesis simulation

```
1  #!/bin/tcsh
2  建議使用絕對路徑
3  vcs -R -error=noMPD \      testbench
4  /home/m123040033/HDL/post_sim/TB.v \
5  /home/m123040033/HDL/behavior/adder_behavior.v \
6  /home/m123040033/HDL/behavior_reg/adder_behavior_reg.v \
7  /home/m123040033/HDL/dataflow/adder_dataflow.v \
8  /home/m123040033/HDL/dataflow_reg/adder_dataflow_reg.v \
9  /home/m123040033/HDL/structure/adder_structure.v \
10 /home/m123040033/HDL/structure_reg/adder_structure_reg.v \
11 /cad/CBDK/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v \
12 +full164 \
13 +access+r +vcs+fsdbon +fsdb+mda +fsdbfile+adder.fsdb +neg_tchk
```

Gate Level Netlist

Standard Cell

Waveform可自行修改檔名

```
adder_behavior behavior(s_behavior, co_behavior, a, b, ci);
adder_behavior_reg behavior_reg(s_behavior_reg, co_behavior_reg, a, b, ci, clk);
adder_dataflow dataflow(s_dataflow, co_dataflow, a, b, ci);
adder_dataflow_reg dataflow_reg(s_dataflow_reg, co_dataflow_reg, a, b, ci, clk);
adder_structure structure(s_structure, co_structure, a, b, ci);
adder_structure_reg structure_reg(s_structure_reg, co_structure_reg, a, b, ci, clk);

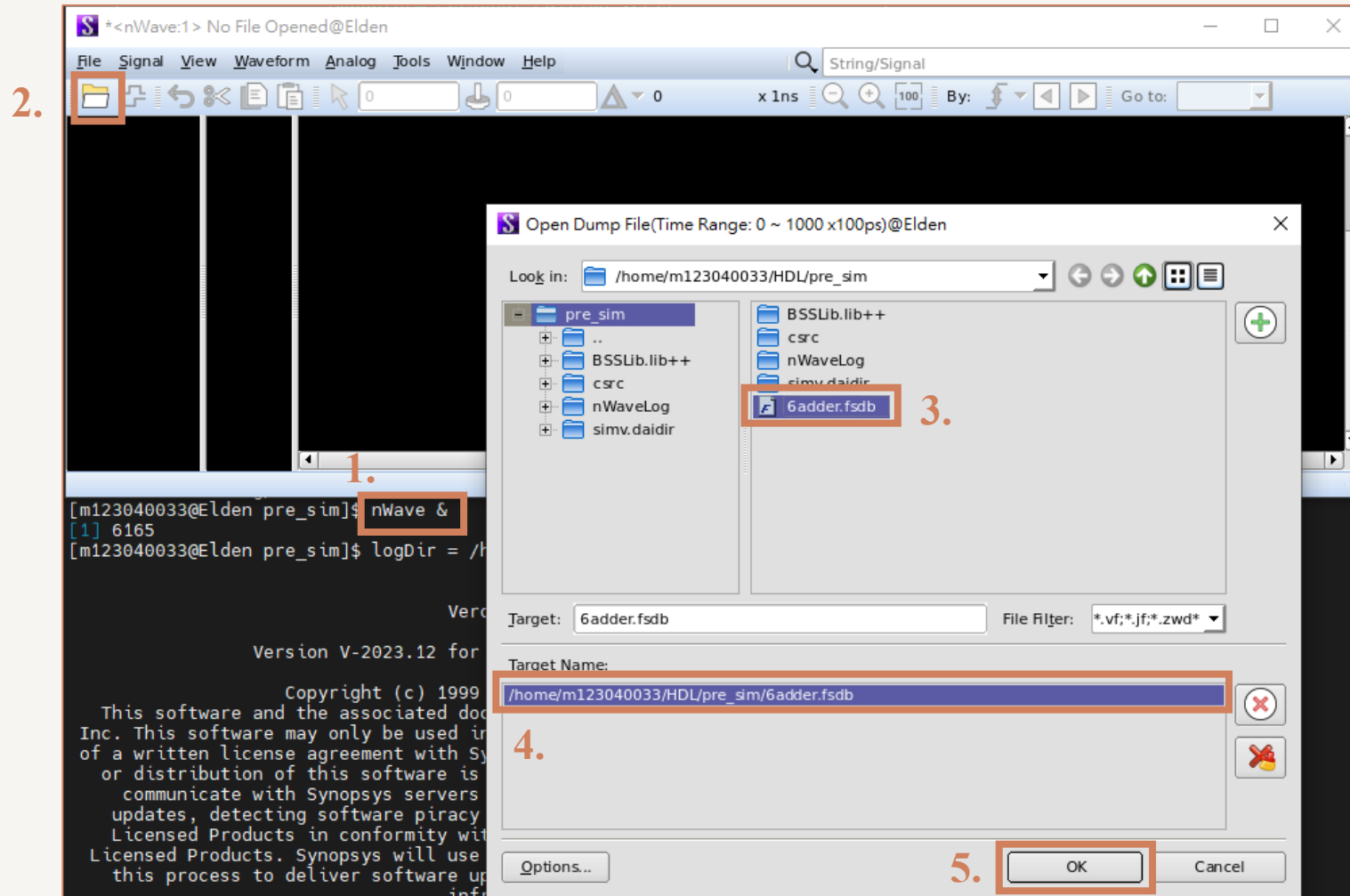
SDF File path 這邊要和呼叫的 Module Instance name 一樣
initial $sdf_annotate("/home/m123040033/HDL/behavior/adder_behavior.sdf", behavior);
initial $sdf_annotate("/home/m123040033/HDL/behavior_reg/adder_behavior_reg.sdf", behavior_reg);
initial $sdf_annotate("/home/m123040033/HDL/dataflow/adder_dataflow.sdf", dataflow);
initial $sdf_annotate("/home/m123040033/HDL/dataflow_reg/adder_dataflow_reg.sdf", dataflow_reg);
initial $sdf_annotate("/home/m123040033/HDL/structure/adder_structure.sdf", structure);
initial $sdf_annotate("/home/m123040033/HDL/structure_reg/adder_structure_reg.sdf", structure_reg);
```

需要在 TB 內加入
Standard Delay Format
(SDF)

nWave (1/3)

- 叫出 nWave : nWave & //&是在背景執行
- 縮放 : Ctrl + 滑鼠滾輪
- 調整訊號位置 : 按住滑鼠中鍵拖曳
- 調整數值表示 : 右鍵 > Set Radix
- 重新執行 VCS 模擬後，不必重開 nWave 和 fsdb : File > Reload

nWave (2/3)



nWave (3/3)

The screenshot shows the nWave software interface with the 'Get Signals@Elden' dialog box open. The dialog box has a title bar with a close button. The main area is divided into three panes:

- Left Pane:** A tree view showing the project hierarchy. The 'HW1_TB(HW1_TB)' folder is selected, and its contents are listed below it.
- Center Pane:** A list of signals available for selection. The signals are grouped by type: behavior, dataflow, and structure. The signals listed are: a[31:0], ans_co, ans_s[31:0], b[31:0], ci, clk, co_behavior, co_behavior_reg, co_dataflow, co_dataflow_reg, co_structure, co_structure_reg, i[31:0], s_behavior[31:0], s_behavior_reg[31:0], s_dataflow[31:0], s_dataflow_reg[31:0], s_structure[31:0], and s_structure_reg[31:0].
- Right Pane:** A list of signals that have been selected for simulation. The signals listed are: a[31:0], b[31:0], ci, ans_co, co_behavior, co_dataflow, co_structure, co_behavior_reg, co_dataflow_reg, co_structure_reg, ans_s[31:0], s_behavior[31:0], s_dataflow[31:0], and s_structure[31:0].

The 'Apply' button at the bottom right of the dialog box is highlighted with a red rectangle. The 'OK' and 'Cancel' buttons are also visible.

1. **Get Signals@Elden**

2. **HW1_TB(HW1_TB)**

3. **s_behavior[31:0]**

4. **Apply**

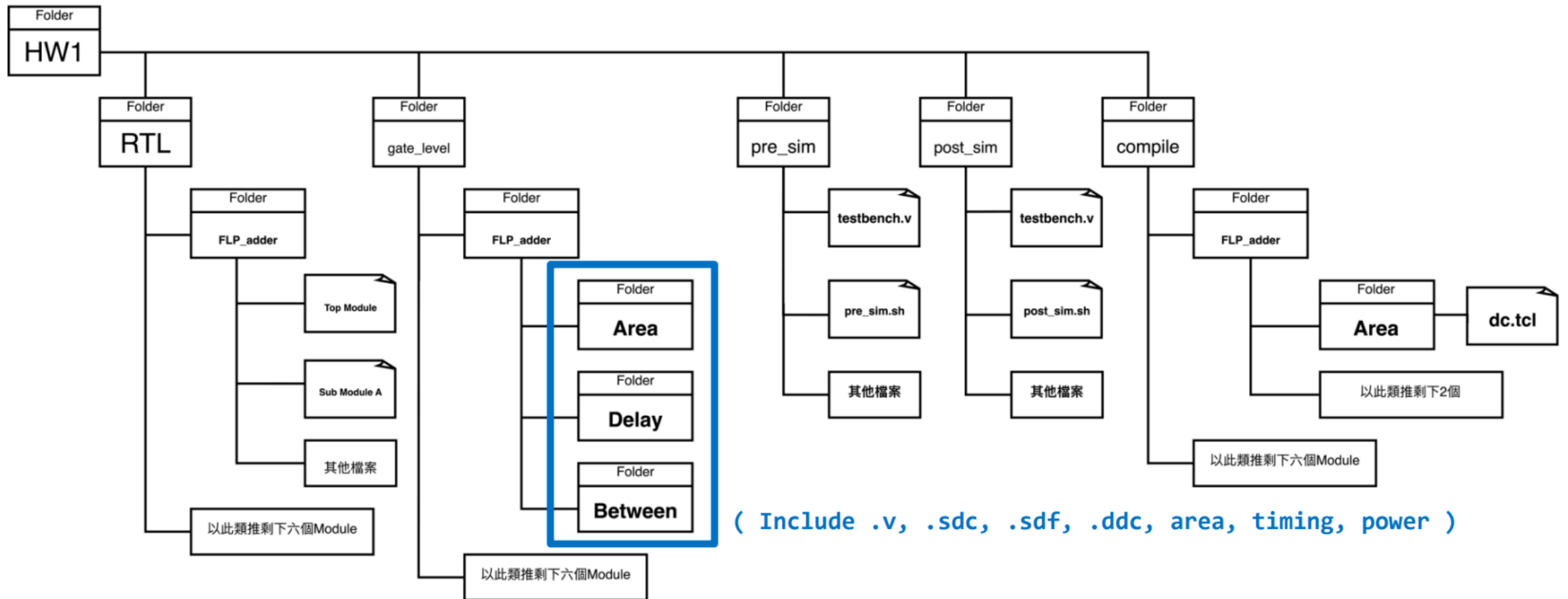
Verdi

有需要的同學請自行到 YouTube 觀看。

<https://www.youtube.com/playlist?list=PLIDduBxjEsW2JBJ1CCH5tOthMWov-kwrx>

<https://youtu.be/wTpzvZs4xZU>

HW1目錄建立方式與流程 (1/2)



HW1 目錄建立方式與流程 (2/2)

1. Write Verilog Code
2. Write Testbench
3. Run Pre Synthesis Simulation – VCS
4. Snapshot Waveform (Pre-sim) – nWave
5. Logic Synthesis – Design Compiler
6. Run Post Synthesis Simulation – VCS
7. Snapshot Waveform (Post-sim) – nWave
8. Complete Report

檔案、top module命名限制

- 作業的top module請按照以下命名規則及input、output順序
fix point版本 RTL, Gate Level 不評分，但需要在報告中與其他電路比較

```
module FLP_adder(a, b, d);  
module FLP_adder(clk, rst, a, b, d);  
module FLP_mul(a, b, d);  
module FLP_mul(clk, rst, a, b, d);
```

- .v檔名請以 <top module name>_<pipeline number>方式命名
(例: FLP_adder_4.v 或 FLP_adder.v)

test bench (1/4)

測資部分格式為 IEEE754 單精度 16 進制的格式，僅供參考 (不可自行減少)

a.txt、b.txt 為 100 筆 Input data，供乘法、加法使用

```
1 558b4567
2 51bc9869
3 5a30dc51
4 3568944a
5 318e1f29
6 3e9b58ba
7 373141f2
8 5ce2a9e3
9 485f007c
10 29200854
11 2116231b
12 2890cde7
13 2a0e0f76
14 281cf92e
15 5fdcc233
16 40a7c4c9
17 46eafb66
18 489b500d
```

IEEE 754 Converter (JavaScript), V0.22

	Sign	Exponent	Mantissa
Value:	+1	2^{44}	1.0880554914474487
Encoded as:	0	171	738663
Binary:	☐	☒ ☐ ☒ ☐ ☒ ☐ ☒ ☒	☐ ☐ ☐ ☒ ☐ ☒ ☒ ☐ ☒ ☐ ☒ ☐ ☒ ☐ ☒ ☒ ☐ ☒ ☒ ☒
Decimal representation	1.91412746322e+13		
Value actually stored in float:	19141274632192		
Error due to conversion:			
Binary Representation	01010101100010110100010101100111		
Hexadecimal Representation	0x558b4567		

test bench (2/4) pre_sim

```
1 // FXP_ADDER / FXP_MUL / FLP_ADDER / FLP_MUL / FLP_ADDER_7 / FLP_ADDER_4 / FLP_MUL_4
2 `define FLP_adder 可以修改來測試各種電路
3
4 // Standard Delay Format File
5 // `define SDFFILE    "./FXP_adder_syn.sdf"
6
7 `timescale 1ns/1ps
8 `define CYCLE 5.0
9 `define DATA_NUM 100
10 `define FILE_A "./a.txt"
11 `define FILE_B "./b.txt"
12 `ifdef FXP_adder
13     `define FXP
14 `elsif FXP_mul
15     `define FXP
16 `elsif FLP_adder
17     `define FLP
18 `elsif FLP_mul
19     `define FLP
20 `elsif FLP_adder_7
21     `define PIPE 7
22     `define adder
23 `elsif FLP_adder_4
24     `define PIPE 4
25     `define adder
26 `elsif FLP_mul_4
27     `define PIPE 4
28     `define mul
29 `endif
30
31 module testbench();
```

```
[2022_alu03@devotion tb]$ ncverilog -f nc_pre_syn.f
ncverilog(64): 15.20-s039: (c) Copyright 1995-2017 Cadence Design Systems, Inc.
Loading snapshot worklib.testbench.v ..... Done
*Verdi* Loading libsscore_ius152.so
ncsim> source /usr/cad/cadence/INCISIV/cur/tools/inca/files/ncsimrc
ncsim> run
-----
-                FLP_ADDER_7                -
-----

All testdata correct!

Avg error in 100 testdata: 5.884856e-09
-----

Simulation complete via $finish(1) at time 5350 NS + 0
./testbench.v:250      $finish;
ncsim> exit
[2022_alu03@devotion tb]$
```

test bench (3/4) post_sim

```
1 // FXP_ADDER / FXP_MUL / FLP_ADDER / FLP_MUL / FLP_ADDER_7 / FLP_ADDER_4 / FLP_MUL_4
2 `define FLP_adder 可以修改來測試各種電路
3
4 // Standard Delay Format File
5 `define SDFFILE "./FXP_adder_syn.sdf" 同時修改成對應 SDF
6
7 `timescale 1ns/1ps
8 `define CYCLE 5.0
9 `define DATA_NUM 100
10 `define FILE_A "./a.txt"
11 `define FILE_B "./b.txt"
12 `ifdef FXP_adder
13     `define FXP
14 `elsif FXP_mul
15     `define FXP
```

```
ncsim> source /usr/cad/cadence/INCISIV/cur/tools/inca/files/ncsimrc
ncsim> run
```

```
-----
-                FLP_MUL                -
-----
```

All testdata correct!

Avg error in 100 testdata: 2.145374e-08

```
-----
Simulation complete via $finish(1) at time 5100 NS + 0
./testbench.v:250      $finish;
ncsim> exit
```

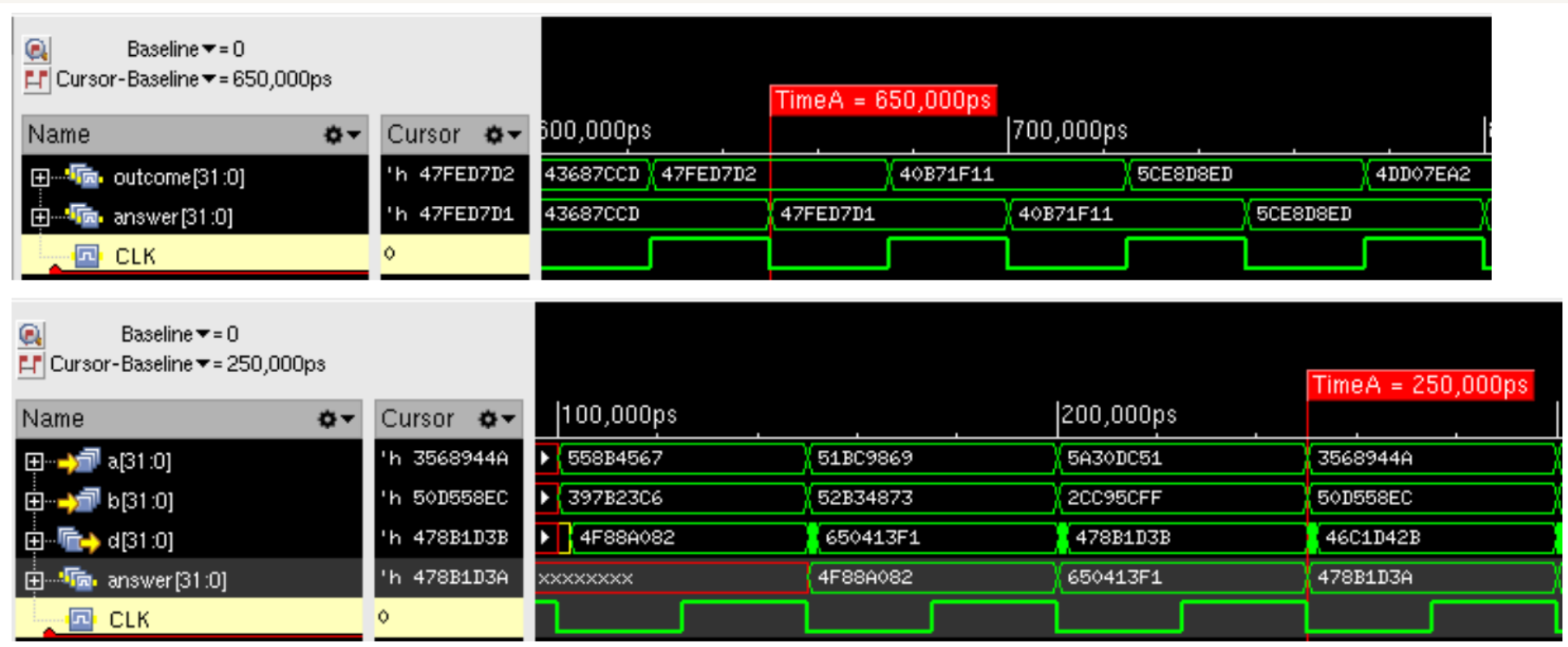
```
`ifdef FXP_adder
    FXP_adder test_module .a(input_a), .b(input_b), .d(outcome));
`elsif FXP_mul
    FXP_mul test_module( .a(input_a), .b(input_b), .d(outcome));
`elsif FLP_adder
    FLP_adder test_module( .a(input_a), .b(input_b), .d(outcome));
`elsif FLP_mul
    FLP_mul test_module( .a(input_a), .b(input_b), .d(outcome));
`elsif FLP_adder_7
    FLP_adder test_module( .a(input_a), .b(input_b), .d(outcome), .rst(RST), .clk(CLK));
`elsif FLP_adder_4
    FLP_adder test_module( .a(input_a), .b(input_b), .d(outcome), .rst(RST), .clk(CLK));
`elsif FLP_mul_4
    FLP_mul test_module( .a(input_a), .b(input_b), .d(outcome), .rst(RST), .clk(CLK));
`endif

        這邊要和呼叫的 Module Instance name 一樣

`ifdef SDF_FILE
    initial $sdf_annotate(`SDF_FILE, test_module;
`endif
```

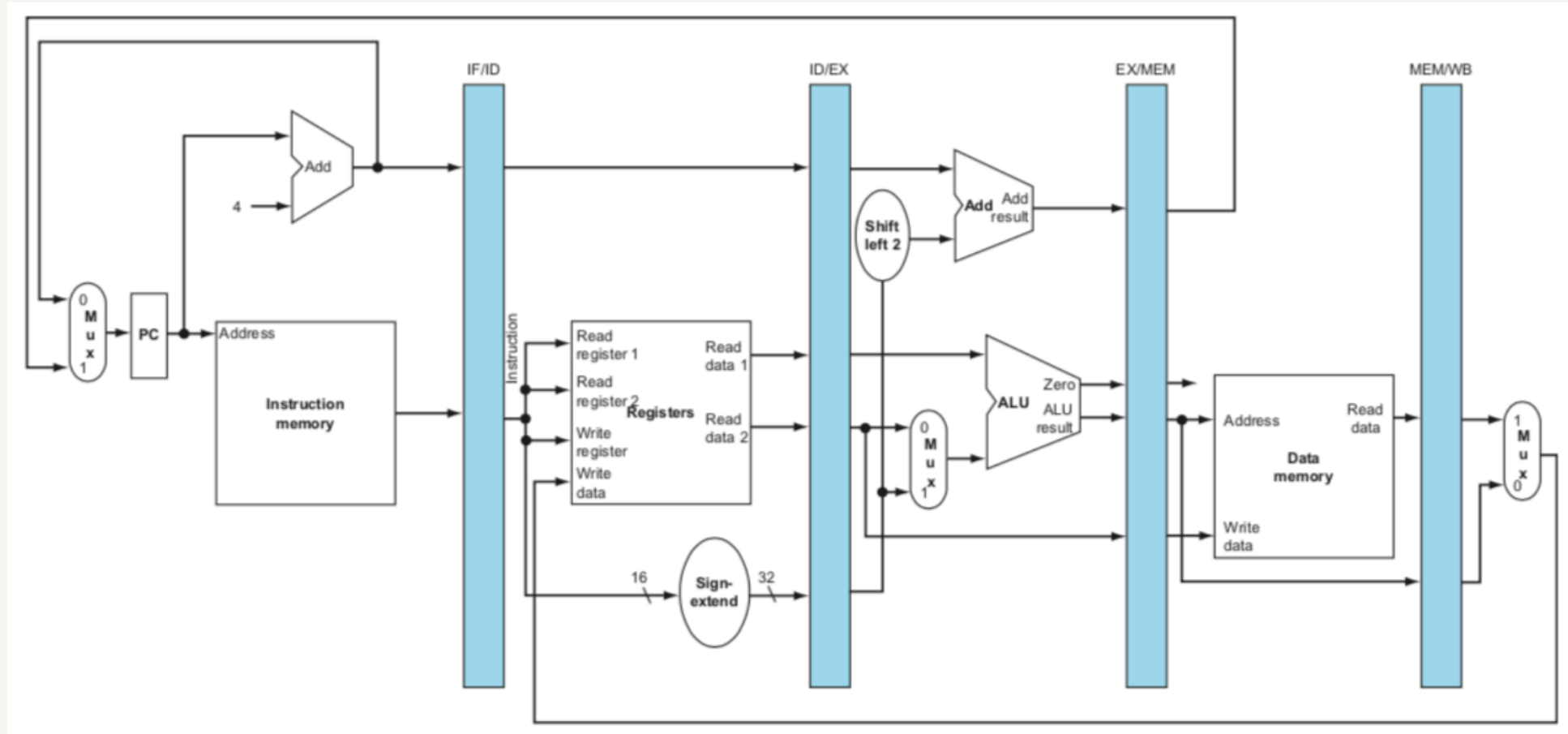
test bench (4/4)

- outcome : 你的硬體輸出
- answer : 軟體計算正確答案



pipeline (1/2)

5 stage = 4 pipeline register, 有需要可以參考 <https://youtu.be/lAnguwdL8K4>



pipeline (2/2)

- dc.tcl 設定教學

#Each Pipeline Delay

```
ungroup -all
```

```
report_timing -from [all_inputs]
```

更改成自己的 Verilog Code 內 Moule Name

可以自訂輸出的 txt 檔名

```
uplevel #0 { report_timing -through { stage1/* } -path full -delay max -nworst 1 -max_paths 1 -significant_digits 2 -sort_by group } > $Path_Syn/timing_report_stage1.txt
uplevel #0 { report_timing -through { stage2/* } -path full -delay max -nworst 1 -max_paths 1 -significant_digits 2 -sort_by group } > $Path_Syn/timing_report_stage2.txt
uplevel #0 { report_timing -through { stage3/* } -path full -delay max -nworst 1 -max_paths 1 -significant_digits 2 -sort_by group } > $Path_Syn/timing_report_stage3.txt
uplevel #0 { report_timing -through { stage4/* } -path full -delay max -nworst 1 -max_paths 1 -significant_digits 2 -sort_by group } > $Path_Syn/timing_report_stage4.txt
report_timing -from [all outputs]
```

```
module stage1 ( a, b, exp_A, exp_B, mant_A, mant_B, sign_A, sign_B );
```

```
input  signed [31:0]a, b;
output sign_A, sign_B;
output [7:0]exp_A, exp_B;
output [27:0]mant_A, mant_B;
wire [27:0] temp_mant_A, temp_mant_B;
wire agtb;
```

```

41 flp_add_pack/add_12/SUM[7] (flp_add_pack_DW01_inc_0) 0.00 0.17 f
42 flp_add_pack/U9/Z (AO22D1BWP16P90LVT) 0.02 0.19 f
43 flp_add_pack/d[30] (flp_add_pack) <- 0.00 0.19 f
44 d[30] (out) 0.00 0.19 f
45 data arrival time 0.19
46
47 各階 pipeline 的 delay 需要填入的數據
48 clock clk (rise edge) 0.25 0.25
49 clock network delay (ideal) 0.00 0.25
50 output external delay 0.00 0.25
51 data required time 0.25
52 -----
53 data required time 0.25
54 data arrival time -0.19
55 -----
56 slack (MET) 0.06

```

Report (1/3)

至少要符合以下所有內容

- 如果有任何硬體測試資料沒有滿分，可以在結果報告說明已完成的部分，和出錯的地方與問題分析，會依狀況部分給分
- 有完成 (All test data correct) 則不須呈現程式碼
- 畫出設計的硬體架構圖並說明
- 在架構圖中明確標示 pipeline 落在哪 (哪些元件為同一階)
- 列出各階 pipeline 的 delay
- 明確標示 pipeline 的 critical path 在哪階
- 列出 delay 最佳化、midpoint、area 最佳化
- 數據表格 – area delay power

Report (2/3)

- 各階 pipeline 的 delay 與 critical path 表格

		Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7	Critical Path	
FLP_Adder_7Stage										
	Delay									
	Area									
	Between									
FLP_Adder_4Stage										
	Delay									
	Area									
	Between									
FLP_Mul_4Stage										
	Delay									
	Area									
	Between									

Report (3/3)

- 數據表格

取消		Area (um^2)			Delay(ns)	Latency	Power (W)		
		CL	SL	Total			Dynamic	Leakage	Total
FXP_Adder	area					(= Delay)			
	delay					(= Delay)			
	between					(= Delay)			
FLP_Adder	area					(= Delay)			
	delay					(= Delay)			
	between					(= Delay)			
FLP_Adder_7_Stage	area					(= Delay * 7)			
	delay					(= Delay * 7)			
	between					(= Delay * 7)			
FLP_Adder_4_Stage	area					(= Delay * 4)			
	delay					(= Delay * 4)			
	between					(= Delay * 4)			
FXP_Multiplier	area					(= Delay)			
	delay					(= Delay)			
	between					(= Delay)			
FLP_Multiplier	area					(= Delay)			
	delay					(= Delay)			
	between					(= Delay)			
FLP_Multiplier_4_Stage	area					(= Delay * 4)			
	delay					(= Delay * 4)			
	between					(= Delay * 4)			

評分標準

- **(40%) RTL code**

8% per module (Fix Point 不評分)

- **(30%) Gate-level module**

2% per module (Fix Point 不評分)

- **(30%) Report (*只收 pdf)**

10% 架構圖 & Pipeline

10% 各階Pipeline數據 (area, delay, mid)

10% 數據表格

繳交方式

1. Server 繳交區

繳交方式：將整個 HW1 資料夾複製到繳交區

在 Server 上執行：`cp -r /MasterClass/<你的帳號>/HW1/ /MasterClass/Homework-Submit/<你的帳號>/`

(例：`cp -r /MasterClass/M143040034_ALU/HW1/ /MasterClass/Homework-Submit/M143040034_ALU/`)

2. 網路大學繳交區

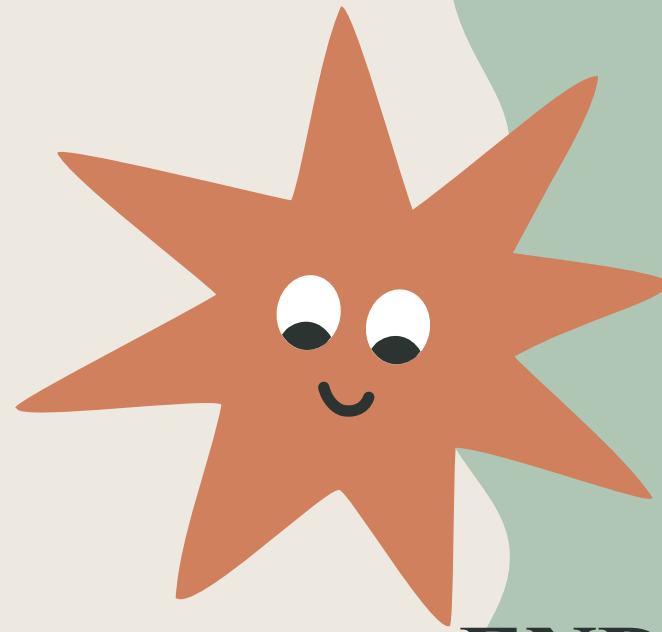
將前面提到 Report 應包含的所有內容加上封面頁

上傳 MXXXXXXXXX_ALU_HW1_Report.pdf 到網大作業區

有任何問題請先寄信詢問

或是提前寄信約 Office hour

E-mail: ina143040034@gmail.com (TA 林以晴)



END

